

School of Computer and Systems Sciences  
Jawaharlal Nehru University  
New Delhi  
Mid-semester Examination  
CS 406 - Numerical Methods  
MCA 1st Semester  
8<sup>th</sup> October 2024

Time: 2 hours

Total Marks: 20

Answer all questions.

Q 1. Find the interval in which the positive roots of the equation  $x - e^{-x} = 0$  lie. Find the smallest positive roots of the equation using Regula-Falsi method. 5

Q 2. Find the solution of the system of equations

$$4x_1 + x_2 + x_3 = 4$$

$$x_1 + 4x_2 - 2x_3 = 4$$

$$3x_1 + 2x_2 - 4x_3 = 6$$

using the Gauss elimination method. 5

Q 3. Using Lagrange's formula, find the interpolating polynomial  $P(x)$  of degree 2 or less that takes the following values. 10

$$x_0 = 1, P(x_0) = 1$$

$$x_1 = 3, P(x_1) = 27$$

$$x_2 = 4, P(x_2) = 64$$

School of Computer and Systems Sciences  
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Mid-semester Examination- II

CS 406 - Numerical Methods  
MCA 1st Semester  
27 November 2024

Time: 2 hrs

Max. Marks: 20

Q 1. Use Newton's forward difference formula to find the unique polynomial  $P(x)$  such that  $P(1) = 1$ ,  $P(2) = 0$ ,  $P(3) = 27$  and  $P(4) = 64$  (5)

Q 2. Compute the following integral using Trapezoidal rule for  $p = 1$ , and number of points = 3 (5)

$$I_p = \int_0^1 \frac{x^p dx}{x^3 + 10}$$

Q 3. Compute approximation to  $y(0.6)$  (4 decimals) of the IVP (10)

$$\begin{aligned} y' &= x(x+y) - 2 \\ y(0) &= 2 \end{aligned}$$

Using Euler's method with step size  $h = 0.2$

School of Computer and Systems Sciences  
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End-semester Examination  
CS 406 - Numerical Methods  
MCA 1st Semester  
12<sup>th</sup> December 2024

Time: 3 hours

Total Marks: 50

Answer all questions. ( $5 \times 10 = 50$  marks)

Q 1. Use Newton's forward difference formula for the given values to find  $f(0.5)$  and  $f(1.5)$

|      |   |   |    |    |     |
|------|---|---|----|----|-----|
| x    | 0 | 1 | 2  | 3  | 4   |
| f(x) | 1 | 7 | 23 | 55 | 109 |

Q 2. Find the solution of the system of equations

$$2x + y + 2z = 10$$

$$x + 2y + z = 8$$

$$3x + y - z = 2$$

using the Gauss-Jordan method.

Q 3. Find the interval in which the positive roots of the equation  $x^2 + 5 = e^x$   $2x - e^x$  lie. Find the smallest positive roots of the equation using Newton-Raphson method.

Q 4. Use the Runge-Kutta method to find approximate values of the solution of the given initial value problem at the points  $x_i = x_0 + ih$ , where  $x_0$  is the point where the initial condition is imposed and  $i = 1, 2$

$$y' = 2x^2 + 3y^2 - 2$$

$$y(2) = 1;$$

Assume the step length  $h = 0.05$ .

Q 5. Compute

$$I = \int_1^2 \frac{1}{x} dx$$

using trapezoidal and Simpson's rules with  $n = 10$ ,  $a = 1$ ,  $b = 2$ . Compare with exact solution.